

SKILL WEEK 7

Task 1:

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {

    pid_t pid;
    int status;

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {

        // Child process
        printf("Child process started.\n");
        printf("Child PID: %d\n", getpid());

        sleep(3);

        printf("Child process completed.\n");

        exit(0);

    } else {

        // Parent process
        printf("Parent process waiting for child...\n");

        waitpid(pid, &status, 0);

        if (WIFEXITED(status)) {
```

```

        printf("Child exited normally.\n");
        printf("Child exit status: %d\n", WEXITSTATUS(status));
    }

    printf("Parent process completed.\n");
}

return 0;
}

```

Task 2:

Code:

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

int main() {

    char *path = getenv("PATH");

    if (path == NULL) {
        printf("PATH variable not found.\n");
        return 1;
    }

    printf("PATH:\n%s\n", path);

    char *copy = strdup(path);
    char *directory = strtok(copy, ":");

    char filename[100];

    printf("Enter executable name to search: ");
    scanf("%99s", filename);

    while (directory != NULL) {

        char fullpath[500];

        snprintf(fullpath, sizeof(fullpath),
            "%s/%s", directory, filename);

        if (access(fullpath, X_OK) == 0) {

```

```

        printf("Executable found: %s\n", fullpath);

        free(copy);
        return 0;
    }

    directory = strtok(NULL, ":");
}

printf("Executable not found in PATH.\n");

free(copy);

return 0;
}

```

Output:

```

[ishivarma@rishis-MacBook-Air Skill week 7 % nano waitpid_demo.c ]
[ishivarma@rishis-MacBook-Air Skill week 7 % gcc waitpid_demo.c -o waitpid_demo ]
[ishivarma@rishis-MacBook-Air Skill week 7 % ./waitpid_demo ]
Parent process waiting for child...
Child process started.
Child PID: 34273
Child process completed.
Child exited normally.
Child exit status: 0
Parent process completed.
[ishivarma@rishis-MacBook-Air Skill week 7 % nano path_search.c ]
[ishivarma@rishis-MacBook-Air Skill week 7 % gcc path_search.c -o path_search ]
[ishivarma@rishis-MacBook-Air Skill week 7 % ./path_search ]
PATH:
/Library/Frameworks/Python.framework/Versions/3.15/bin:/Library/Frameworks/Python.framework/Versions/3.14/bin:/opt/homebrew/bin:/opt/homebrew/sbin:/usr/local/bin:/System/Cryptexes/App/usr/bin:/usr/bin:/bin:/usr/sbin:/sbin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/local/bin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/bin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/appleinternal/bin:/pkg/env/global/bin:/Library/A
pple/usr/bin

Enter executable name to search: ls
Executable found: /bin/ls

```